

199

EXPONENTIAL GROWTH

Lincoln University's
Computing Story to 1990

Neil Mountier

Centre for Computing and Biometrics
Lincoln University

Dedication

Clifford Vaughan Collister 1954-1990

Cliff Collister joined the Computer Centre at Lincoln in 1979 after graduating with an M.Sc. in Computer Science from Canterbury University. He had an unusual combination of talents. Very intelligent with a flair for solving all kinds of technical problems, he also had a real interest in helping the less-skilled users. He was enthusiastic about demonstrating computing to school parties and to other visitors. Cliff died in an avalanche on Mt Olympus on 11 August 1990, at the end of a skiing holiday. His warm heart and keen mind were a great gift to the people he worked among.

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TABLE OF CONTENTS

Prefatory Note	ii
Chapter 1 Beginnings - the 60s	1
Chapter 2 A Frustrating Decade - the 70s	6
Chapter 3 Real Progress - the 80s	12
Chapter 4 The Whole Period - Exponential Growth	27
APPENDIX 1 Staff employed in the Computer Centre and the CCB	38
APPENDIX 2 A description of some earlier equipment ...	40
APPENDIX 3 Lincoln Campus Network in December 1989	42

Photographs appear between pages 20 and 21.

Prefatory note

Lincoln University began as the Lincoln School of Agriculture in 1878, to teach scientific and practical farming. Its functions have expanded since then, and its name has changed more than once, too. For many years it was Canterbury Agricultural College, a constituent college of the University of New Zealand. In 1961, when the University of New Zealand was abolished and separate institutions established, Lincoln College retained its own Council and Professorial Board and its separate funding, but became formally a College of the University of Canterbury. That formal status was changed in 1990 when it assumed the title of Lincoln University.

There are now about two thousand students at the University. The scope of its teaching has widened over the years, and its Charter highlights four broad areas of teaching and research: commerce and management; primary production and natural resources; science and engineering; social sciences. Computing plays an essential part in all these areas.

Chapter 1

Beginnings - the 60s

In August 1967 an IBM 1130 computer was installed in the still uncompleted Hilgendorf Building. The Hilgendorf Wing, as it is sometimes described, was the first of the major teaching and research buildings that have so greatly changed the appearance of the campus. And the computer was the first resident of the second floor of that building, along with a newly appointed computer programmer, Esmée Emerson. The director of the Computer Centre still had his office in a 'tin hut' some distance away. This was the first computer on the Lincoln campus, and it came relatively early in the New Zealand experience of computing. The first university computer installation in NZ was at the Ilam campus of Canterbury University where an IBM 1620 had been installed in May 1962, not long after the first computer in NZ the Treasury IBM 650 which began work late in 1960.

Of course, the arrival of this first Lincoln computer was the result of a process of some years. Members of Lincoln staff, mainly from economics and farm management, had made use of the facility at Ilam from the year of its installation. Bryan Philpott was in charge of Economics and he had Alastair McArthur, another innovative thinker on his staff. Mary Matheson was on the staff of the Agricultural Economics Research Unit, also under Philpott's direction, and all these people became early users of the Ilam computer. But computer facilities have always become overloaded, and when this happened at Ilam, Lincoln staff were threatened with the withdrawal of their access,

and this had led to the proposal to procure a computer for Lincoln itself. In May 1965 a meeting was held to discuss this proposal. The Professors of Farm Management, Agricultural Economics, and Agricultural Engineering were at this meeting and Bruce Moon, Director of the University of Canterbury Computer Centre attended in an advisory capacity. Following this meeting, applications were made to outside bodies for funding, but nothing was forthcoming.

A decision was made in 1966 to hire an IBM 1130 with an option to purchase at a later date. Deliveries of computers at that time were very slow, and mid-1968 was promised. But then another IBM customer cancelled an order and Lincoln was offered the earlier delivery time of August 1967. The option to purchase was, in fact, taken up a year later at a cost of \$55,000. The funds for this were found within the institution apart from a grant of \$15,000 from the Research Committee of the University Grants Committee.

To help put this in the context of the time, it could be noted that Auckland University installed its first computer just a few months earlier in 1967 and it was also an IBM 1130. Canterbury University dealt with the overloading on their machine by installing in October 1967 a much larger computer, an unusual member of the IBM 360 series, the 360/44.

The computing experience in the 1960s

The first uses of the Lincoln computer were for research, simply transferring the work that had been being done at Ilam. Of course, the proximity of the machine encouraged users to do more computing and also kindled the interest of new users. To grasp something of the effect that this proximity had, it is necessary to know how computing was experienced by the user in those days.

The computer user did not have commercially produced 'packages' to use. There were a few programs supplied by IBM which were of very limited scope, and there were some sets of sub-routines for programmers to use. Otherwise users were dependent on their own

resources. The user generally wrote his/her own programs or in a few cases used programs written by a member of the same department or by the computer centre programmer. There were no interactive terminals; communication with the computer was by the 80-column punched card, which was prepared on a different machine and then fed into the computer. Both program and data were entered in this way, and when (rather than if) an error was found, the offending card(s) had to be found and replacement cards punched and then the whole deck resubmitted to the computer. This process was repeated until the program appeared to work correctly. Results were printed on a line-printer in the computer room. The line-printer printed only numbers and upper-case letters, and deciphering the output was not the most straightforward of tasks. During the working day, a member of the Computer Centre staff would look after the input of card decks and the collection of output from the printer. In evenings and at weekends, suitably trained users would run their own jobs through the machine. Writing and getting even a simple program to work was a task that was spread over several days at best, and more often weeks. Only the keen were users.

However, it is not hard to see why a computer on campus was much more attractive than one 14 km away. The time taken to get a program working was reduced considerably. Short courses for potential users were run, and computing began to be an accepted part of life for some people. In 1968 an additional staff position was created for a punch-card operator, who could produce correctly punched cards for users much more efficiently than they could do the job themselves. Ruth Arnst was the operator from 1968 till 1972, and she quickly became an operator of the computer as well as the producer of punch-cards.

Administration

Neil Mountier had been appointed Director of the Computer Centre in addition to his existing duties as senior lecturer in biometrics. He had been previously on the staff of the Department of Agriculture in Wellington and had made use of the IBM 650 at Treasury from its installation in 1960. Although he was attached to the Plant Science

Department, the computing organization was set up as an independent entity from the outset, with the Director responsible to the Principal. A committee of the Professorial Board was set up to guide policy in computing, with the Director as an ex officio member. This had Professor J.D. Stewart as its Chairman from 1967 till 1974.

This proved to be a very sound administrative basis for a computer centre. It was important in the initial stages that it was not identified with a particular teaching department, and that the Committee allowed input from, and returned information to, interested departments. This form of organization has been expanded and developed, but not basically changed since the outset.

Teaching

It soon became apparent to some that our students would benefit from a formal course in computing, and after some discussions, Professorial Board agreed that a course could be offered as a half-paper to Masters students. This course was called Computer Methods and aimed to provide understanding and skills to those students who expected to make use of the computer in their research for their theses. The course consisted of lectures plus practical work on programming in FORTRAN, on general computer understanding, on elementary numerical methods and on an introduction to simulation methods. Neil Mountier and Alastair McArthur shared the teaching of the course for many years with some input from other staff. The subject proved popular from the outset. The first class in 1969 had 26 students begin and 18 complete the course. Part of the requirement of the course was the completion of an individual computing project, preferably related to the student's area of research. Project topics covered a wide range. In that first year they included simulation studies on market gardening, on tractor replacement policy and on dam size. Other topics included evapotranspiration calculations, generalised matrix inversion and a perpetual calendar. Students of the course all had some hands-on experience with operating the computer, and some acquired much more when they came to process data for their theses.

The IBM 1130 was an up-to-date computer in the 1960s, but its speed and capacity were considerably less than that of the commonly used personal computers of the late 1980s. Its main memory, for instance, was only 16KB, compared with the 640KB that is the minimum on most PCs. Consequently, some computing tasks took a long time, and some research students would operate the computer in evenings and at weekends to get their work done. It was not uncommon for students to work right through the night. Of course, nothing lengthens a computing job more than inefficient programming, and there were some startling cases of that. But simulation is a method that commonly requires a lot of computer time, and some student work was quite ambitious.

For a few years, Computer Methods was the only credit course offered to students, although short courses on programming were given to staff and students from time to time. It was not until the Agricultural Commerce degree was introduced early in the 1970s that an undergraduate computing course was offered at Lincoln.

Chapter 2

A Frustrating Decade - the 70s

In the 1960s, each university had gone into computing on its own, asking the UGC for financial assistance where it could. By the end of the 60s most universities were looking at improving their equipment, but were realising that it would be very expensive to get what they needed. There was an idea in the air that one or two large computers might serve all the universities more effectively and with less cost than seven smaller ones. With a new funding quinquennium imminent, the Government of the day proposed a scheme to enhance university computing facilities throughout the country by providing a capital sum (\$3,350,000) for equipment and asking the universities to agree amongst themselves on how and where it should be spent. (There were a few cynics about who suggested that the Government were confident that their money was safe, because the seven university institutions would never manage to agree). The Vice-Chancellors' Committee responded to the government's challenge by setting up in June 1969 a sub-committee with one representative from each of the seven institutions under the chairmanship of Dr. Robin Williams, the Vice-Chancellor of Otago University. The representative was generally the director of the computer centre or the chairman of the computer committee. This sub-committee eventually took on a permanent form as the Vice-Chancellors' Standing Committee on Computers.

This committee first sent a small group of its members to look at what was being done in the USA and Great Britain, and then invited proposals from computer firms to meet the perceived needs of the

universities. In a week in December 1970 the committee met in Wellington to hear presentations of these proposals from firms. The result of these deliberations was a contract with Burroughs to supply five B6700 computers together with two DC1200s. The B6700s were moderate sized computers and were to be situated on each of the campuses except for the two smallest, Waikato and Lincoln. The latter received the DC1200s connected by line to Auckland and Canterbury respectively. These DC1200s were small computers equipped with card-readers and line printers, and had the task of managing communications down a telephone line to the host computer. They were known as RJE terminals, where RJE stood for Remote Job Entry.

Although the recommendation of the sub-committee was quickly accepted by the Vice-Chancellor's Committee, some subsequent politicking by two other computer firms led to a 6-months delay in formal ordering. Delivery times were still quite lengthy, and the Canterbury B6700 was not installed until December 1972, and the Lincoln DC1200 eventually was put through its acceptance testing in August 1973.

Lincoln-Canterbury cooperation

The arrangements for equipment meant that Lincoln had a share in the resource that was the B6700 computer on the Ilam campus. This in turn implied that Lincoln was responsible for a share of the cost of running the B6700, as well as having the cost of maintaining the DC1200 which simply acted as an access to the Ilam computer. Of course, this cooperation needed some administrative structure and this was provided partly by cross-representation on each other's computer committee, and partly by a high-level joint committee which met infrequently.

There was a lot of good-will between the two computer centres, and much effort was put into getting the system to work. However, the communications software did not perform to expectation and although some useful work was done by Lincoln users, the service was never satisfactory. Waikato University were having similar problems with

their link to Auckland, and a joint meeting was held in September 1975 at which it was agreed to replace the unsatisfactory DC1200s with equipment from Digital Equipment Corporation which had the potential to function both as a communications link to the B6700 and also provide some stand-alone computing capability so that the aging 1130s could be retired. A Digital PDP11/40 was delivered to Lincoln in June 1976 and by early August had taken over the running of the RJE link to Ilam. It used the RT11 operating system with communications software developed by Canterbury staff. This proved to be a great improvement. Our own programming staff at that time consisted of George Gregg, an experienced programmer from California who had come to New Zealand seeking a better way of life, and Martyn Brown who had joined the Computer Centre at the end of 1972 when he graduated from Canterbury University. With communications to the B6700 running smoothly, they set about developing software to allow the PDP computer to act simultaneously as a local computer and a communications link to Ilam. This required the more complex RSX11 operating system, but it promised a much better service to our users.

Computing in the 1970s

At this time in the mid-1970s, using a computer at Lincoln was still a very similar process to that described earlier: input was still by punched card and output was on a line printer in the computer room. The interactive terminal was in use in some parts of the world (for instance, at CSIRO in Australia) but it was still a rarity in the NZ university system. The B6700s that the NZ universities had purchased were of minimum configurations, and the additional large sums of money needed to expand them sufficiently to support an interactive service were not provided. The great majority of work was done in batch mode with card input and line-printer output. One thing that was changing was the use of commercially supplied program packages. One of the first of these to achieve wide use was SPSS, which stands for Statistical Package for the Social Sciences. Such packages needed large computers and access to these was one of the main advantages for Lincoln of the link to the B6700, and a number of Lincoln people used SPSS in this way. Another package used at this time implemented the

operational research technique of Linear Programming (a term that has no relationship to computer programming). This technique was used extensively by Farm Management staff and a proprietary Burroughs package called Tempo met their needs.

Another feature of the computing experience of that time was the frequency of system failure. It would be a very unusual week in which a system failure of some kind did not occur, and on many occasions there would be more than one failure in a day. One of the hazards of operating at a distance was that there were more things to go wrong. Users at Ilam were subject to the failure of hardware on the B6700 and also to a 'crash' of the operating system, both quite common occurrences. Lincoln users suffered all the effects of these problems. In addition they had to contend with hardware failure in the Lincoln equipment, problems with the telephone line joining the two computers, problems in the communications software and crashes in the operating system on the PDP computer. It was an uncertain environment for the computer user.

One notable change that occurred in 1972 was the construction of the Computer Centre building adjacent to the Hilgendorf building. This was built to house the 1130 and the DC1200 and to provide air-conditioned space for them both. It was enlarged in 1985 to provide additional teaching laboratory space.

For much of this decade Lincoln was getting a very limited computing service and paying quite heavily for it, because the payments to Canterbury University took nearly half the Lincoln computing budget. The IBM 1130 gave good service for 10 years, and was retired in 1977 when it was no longer economic to run because a better local computing capability was available on the PDP11/40. The software task of providing local computing simultaneously with communication to Ilam was a much larger one than was initially realised. Our two programmers worked very hard and succeeded in their task. But, of course, as soon as something is running, enhancements are asked for, so that the task was never really finished. However, by 1980 the Lincoln computer user was getting a more tolerable level of computer

service, and a few interactive terminals were at last available for local computing. The ability of the PDP11 to provide interactive access made possible the first step in the profound change from the batch use described earlier to the mode of use that is normal today. Another change that affected some users was the provision of 'floppy' disk drives. Instead of keeping large data files as packs of cards to be read in each time they were needed, these files could be stored on an inexpensive flexible disk and read from this much more quickly. Another facility that was in use by the end of the decade was a graph plotter (installed in 1978) which quickly became popular with students preparing figures for theses. Altogether, things were moving positively but users, as well as Centre staff, were still considerably frustrated by what could be done on the combination of a small local computer and a distant, larger machine. Computer Centre staff were kept so busy tending the complex software that provided for these operations that they had very little time to assist users.

Teaching

Lincoln degrees were almost entirely prescribed at this time. The only choice that a student could exercise over the subjects taken was normally confined to the final year. This situation is normal enough in a professional school, and there are good reasons for it. However, it does make the introduction of new areas of study quite difficult. A new subject can enter a course only if an existing one is dropped. Since lecturers generally believe that the subject that they are presenting is of vital importance to the future career of all their students (and they would not be very good lecturers without that enthusiasm), it requires a great deal of pressure to get a new subject into such a degree. Even when a large number of teaching staff see the importance of the new subject, they will still not be willing to give up any of their own teaching time to accommodate it. It can be a long and difficult struggle to introduce a new paper into a professional degree.

In the early 1970s, computing was regarded by many on the Lincoln campus as irrelevant to agriculture. But it was increasingly obvious to a few that computing would loom large in the working lives of many of

our students, and that it would be a neglect of duty to send them out into the working world without any familiarity with computer techniques and their potential.

What was done was what could be managed in that environment. The most important change was the introduction of computing to the new degree in Agricultural Commerce. This had two streams, and Bryan Philpott agreed to include in the economics stream a Computer Science paper as a compulsory element in the second year. It was offered first in 1972 to a class of 24 students, and set out to be a course similar to those offered in other universities at that time. It had a heavy emphasis on programming. The Science degrees received a more modest input. In the second year of the B.Ag.Sci. and B.Hort.Sci. degrees there was a course in Biometrics, which was under the control of the Computer Centre Director wearing a different hat. A section on Computer Acquaintance was inserted into this subject from 1969. This was modest in its aims as the title would suggest, and occupied just a dozen lectures. For several years this was the only computing available to science undergraduates, but an optional third-year subject called Biological Science, which was run as an inter-disciplinary venture, provided an opportunity in which some applied computing could be used in modelling and in growth analysis.

An undergraduate diploma course designed to train field technicians was established in 1975, after discussions with MAF and DSIR. Neil Mountier and Dick Lucas designed the diploma's central course on Experimentation. As a common task of many field technicians was dealing with the detailed figures that experiments produce, a familiarity with computational methods was part of this course. The emphasis at first was on the use of a desk calculator, but this changed with time to the use of a computer and its software. The inclusion of computing here could be managed with ease because of the dual role of the Centre director.

Chapter 3

Real Progress - the 80s

As the 1980s approached, the University Grants Committee received submissions for the next funding quinquennium, and these submissions contained requests from every institution for computing funds to replace the now obsolescent equipment that was purchased at the beginning of the 1970s. It was obvious to all people involved in computing that there was a need for major funding, and apparently this was confirmed by other advisors of Government. Certainly, the UGC convened a meeting of computing representatives from each of the university institutions in mid-1980. That first meeting gave a real scare to the representatives of Lincoln and Waikato, when it appeared that the interpretation being used by the meeting's chairman (Jack Caldwell of the UGC) was that funds would be divided among only the five sites with B6700s. However, in the following weeks some intensive lobbying was done. Other university representatives displayed a sense of fairness, and at the second meeting it was accepted that all seven institutions were to receive funding for new computing equipment.

Negotiations for the supply of equipment were handled very formally by the UGC, tenders being called for on the basis of specifications submitted by the universities, and with direct contact between tenderers and computer centre staff strictly forbidden. Tenders were received in October 1980 and a meeting was arranged at which companies tendering could expound on their offerings and at which universities could question them on details. This whole process contrasted with the kind of negotiation that computer centre directors were accustomed to when acting for an individual university. There was usually a great deal of discussion between the university and the

supplier to find the most effective configuration of equipment for the funds available, as well as a certain amount of haggling over prices. However, there was a satisfactory outcome for all universities from the tendering process, and for Lincoln in particular it represented a great leap forward.

The time from ordering to delivery was much shorter than it had been ten years earlier. Orders were placed in December 1980 and in April 1981 a Digital VAX11/780 with sufficient memory and just enough disk storage (see Appendix 2) was installed at Lincoln. This machine could run all standard software packages, and the expensive link to Ilam could be dropped. The VAX also ran interactive terminals well. At the time of installation 25 terminals were available and this soon increased to 50. The small dot-matrix printer was now available and three of these were also installed. For the first time computer printing facilities were available outside the computer room. These printers could be tended by users, and therefore could be placed at convenient places around the campus. Some of us had thought that computer users tended to be conservative about change and that many would take a time to change from the card-input/line-printer output mode that they had been accustomed to. But the convenience of the interactive terminal where output can be immediate on the terminal screen, and almost as quick on a printer nearby was so overwhelming that the use of the punched card shrank to almost zero within a few weeks of the VAX being installed. From this time, the way in which a computer user operates at Lincoln has not changed fundamentally. An integrated keyboard and screen and ready access to hard copy facilities are things users take for granted today. They are a very different experience from that of the 1960s and 70s. Subsequent changes during the 1980s have been important and considerable, but there has been no change in the users' basic mode of operation such as occurred in 1981.

George Gregg, who had laboured so ably in the difficult years of the 70s, left at the beginning of 1981 just before the VAX was installed. He was replaced by Alan McKinnon, an electrical engineer by training who had moved into computing some years earlier. Martyn Brown was still there, and the staff had been added to in 1979 by Cliff Collister, another Canterbury computer science graduate. In July 1981 another programmer was appointed, this time a UK graduate named Andy Griffiths. This capable team with good hardware and a sound

operating system (VMS) provided a computing service that was better by an order of magnitude than anything Lincoln users had experienced before.

Administration

The administrative structure set up in 1967 continued through the 1970s. Barry Dent, who succeeded Jim Stewart as Professor of Farm Management, became chairman of the Computer Committee in 1974 and remained in that position until he left Lincoln in 1986. There were some adjustments to cope with the link to Canterbury University in the 1970s, but otherwise the relationship between the Committee responsible for policy and the Centre responsible for carrying out the policy went smoothly. The good relationship between Committee and Centre was an important factor in the achievement of computing service.

In 1983 the Centre for Computing and Biometrics was established. This brought together the lecturing staff, who had previously been under the umbrella of the Plant Science Department, and the staff of the Computer Centre under the directorship of Neil Mountier. This seemed a natural development from the growth that had occurred. The staff numbered 13 at the time of establishment, made up of 4 academics, 5 programming staff, two and a half operators, a secretary shared with another department and a director. By 1989 this had grown to 20. Most universities have separated the teaching of computing from the running of a computer centre, but Lincoln's small size has made their combination in the CCB a fruitful arrangement.

Equipment developments

The installation of the VAX 11/780 marked the beginning of a new era in computing at Lincoln because it provided on campus virtually all computing facilities that could be available at the time. It was a basis on which to develop. It was supplemented by another, smaller VAX 11/750 in February 1984, and these two machines were linked in the computer room, so that it was possible to access either one from any

terminal. The linkage was by a form of coaxial cable known as Ethernet, the first example of such a connection in New Zealand. Later in the decade (January 1987) a campus-wide Ethernet network was set up with coaxial cable linking the main buildings with a much more sophisticated communications scheme. At the end of 1989, there were over 230 terminals and workstations connected to this network.

Laser printers were a development of the 1980s, and they offered a quality of output unobtainable earlier. The first laser printer at Lincoln was installed in 1985 and quickly became popular despite the higher costs of production. This type of printer could produce graphical output of good quality in black and white and could use a wide range of font sizes and types. 'Desktop publishing' became possible with a laser printer when some quite sophisticated software was used. The production of theses for advanced degrees has been greatly helped by this kind of printer.

A more recent acquisition was a scanner which could 'read' a picture, or a map or anything else on a page into the memory of a computer. The computer could then scale the picture up or down and it was available to be printed out again on a laser printer as a part, or the whole, of a page.

Off-campus communications

The first communications link outside the campus was the connection to the B6700 computer at Canterbury University in the 70s referred to in the previous chapter. This was by a leased telephone line. When the Lincoln VAX was installed in 1981, both the Ministry of Agriculture and the Wool Research Organization were interested in having some access to it. At that time the MAF had temporary premises on campus, and a local line was run to provide them with a terminal connection. Wool Research were across a public highway, and the law required the use of a Post Office telephone line for their connection. A line was accordingly leased.

A different feature was the provision of an auto-answer modem to the VAX in 1982. This allowed a staff member to take home a terminal and modem which was connected to a telephone receiver by a rather clumsy acoustic coupler. It was then possible to dial up the VAX and work away from home. This was a slow line (300 bits per second), so that response time was sometimes frustrating, but the facility was much appreciated by some people. In 1987 this was replaced by a faster multi-speed modem capable of up to eight times the speed.

By 1985 the Agricultural Engineering Institute on campus had a need for data communication with an operation they ran in Hamilton. By that time Telecom had established a packet switching service known as PACNET. This allowed the transmission of digital data to other users of this system, without having the expense of leasing a telephone line for the whole distance or, alternatively, paying standard toll charges for a slow-speed line. There were some appreciable costs to joining this system, but an agreement was reached on funding between the Centre, the AEI, Wool Research and the Library, all of whom were wanting this type of connection. The original character-mode connection to PACNET made in 1985 was upgraded in 1986 to a packet-mode connection which allowed for several simultaneous users.

For some time the New Zealand university computer centres had been considering better means of communication between them. At a meeting in 1985 it was agreed to establish this on PACNET connections, and in 1986 this was put into effect. It was then possible to send electronic mail to any other NZ university and also to overseas university and research establishments. A gateway to North America was provided at Waikato University. The overseas communications have since been upgraded to a permanently leased circuit to Hawaii. Many individual research workers have found this of great benefit when collaborating with colleagues in other places.

PACNET provides a good service for mail, but is not economic for the transfer of large files or the remote use of a computer at another site. As demand for such services grew, the NZ universities moved towards a leased-line system of their own. Of wider significance was the

agreement in 1989 to set up a leased line network between NZ universities, DSIR and MAF which is known as TUIA, which means 'together'. This will further benefit research cooperation in New Zealand.

The Personal Computer

Perhaps the most striking computing development of the decade was the rise of the personal computer or PC. Although the first personal computers were marketed in the latter part of the 70s (the Apple II from 1977), by 1980 the PC was just in its infancy. By the end of 1989 it was commonplace. During that time its power and versatility have increased greatly, and its price declined markedly. It is used for word-processing, for accounting in small businesses, for publishing, for design work in engineering and architecture, for many kinds of scientific purposes. Its power has continued to increase and with it its scope.

It is another instance of the extraordinary technological development that has been such a feature of electronic computing throughout its short history. When small machines were first made, they looked like toys to most people. Indeed, a major use of the early examples was for playing games. And those 8-bit machines, many of which used the operating system known as CP/M, were rather limited in their scope. Nevertheless, Lincoln had advocates and users of these machines. Several found their home in the Burns Wing in the units associated with the Farm Management and Economics departments. Professor Barry Dent voiced the need for students in Farm Management to become familiar with the software that was being written for these CP/M machines. But it was clear that the scene was changing rapidly with the advent of 16-bit machines. In 1984 the Computer Centre found an interim solution to the teaching need by acquiring some devices that allowed VAX terminals to perform as CP/M machines. These small boxes were designed and built at Massey University at a low price, and they achieved their end by having the VAX act as a disk system to the CP/M box.

The PC really got moving when IBM produced its PC in 1981. IBM have dominated the computer business since the mid 1950s and for much of that time have had more than 70% of market-share. So when IBM produced a PC it quickly became the standard. Other manufacturers produced cheaper look-alikes that used the same operating system MS-DOS. Software of all kinds was written to work under this operating system, and the big handicap of the earlier PCs (that you could not do much with them without writing your own programs) gradually disappeared. In 1984, Apple produced the Macintosh which broke new ground in its graphics capability and 'user-friendliness'. By now the real potential of the personal computer was beginning to emerge. It is worth noting that even then those at the centre of things had a limited view. In the mid-70s, two major computer firms, Digital and Hewlett-Packard passed by opportunities to market personal computers on the grounds that no one would want to use them.¹ Microsoft, the largest software firm in the world provided the operating system, and wrote into it a limitation on memory size that of recent years has bugged many people. They were sure then that nobody would possibly want to use more than 640KB of memory. By late in the 80s it was possible to buy personal computers with six times that amount of memory.

An area the personal computer greatly influenced from early on was word-processing. The day of the typewriter was seen clearly to be passing as the advantages of typing by computer became evident. Although there was some resistance, as there always is to new technology, the ease of making corrections and changes, the access to a variety of printers and the ability to move a paragraph to another part of a document were just a few of the reasons that word-processors were attractive. At Lincoln a decision was made in 1985 to provide departmental secretaries with word-processors, and after a good deal of investigation a set of 14 Rainbow PCs running the MASS-11 word-processor was provided. A printer was provided with each, and access was available also to a laser printer in the Computer Centre for high

1. Stan Augarten in 'Bit by Bit' 1984

quality print. The number of machines was subsequently increased, and in 1986 a change was begun to IBM look-alikes, but running the same software.

By 1985 a new teaching laboratory was being planned, and there was a choice between setting it up with a network of PCs or with terminals connected to a central processor. Network software was still in an early stage of development, and the second option was chosen. However the progress of the PC and of networking continued apace and in January 1987 a laboratory of networked PCs was installed in the ground floor of the Computer Centre building. By the end of 1989 there were about 220 personal computers on campus.

Administrative Computing

The Registry had acquired a small Burroughs computer (a B700) in 1975. A staff position was also allocated and Bill Little was the first appointee. He began the task of gradually adapting financial systems to run on the computer. However, he stayed for only a little over a year, and was replaced in 1976 by Ian Johnson who continued the work over the next few years. The academic section under John Scott had already made some forays into computing on Computer Centre equipment, but progress was slow as the Computer Centre was not staffed to provide the amount of help needed, and Ian Johnson was fully occupied with the finance area. By the time of the VAX installation it was clear that all Registry work would be handled more effectively on the VAX than on a small stand-alone machine. Interactive terminals and a printer were placed in the Registry, and preparations for transfer begun.

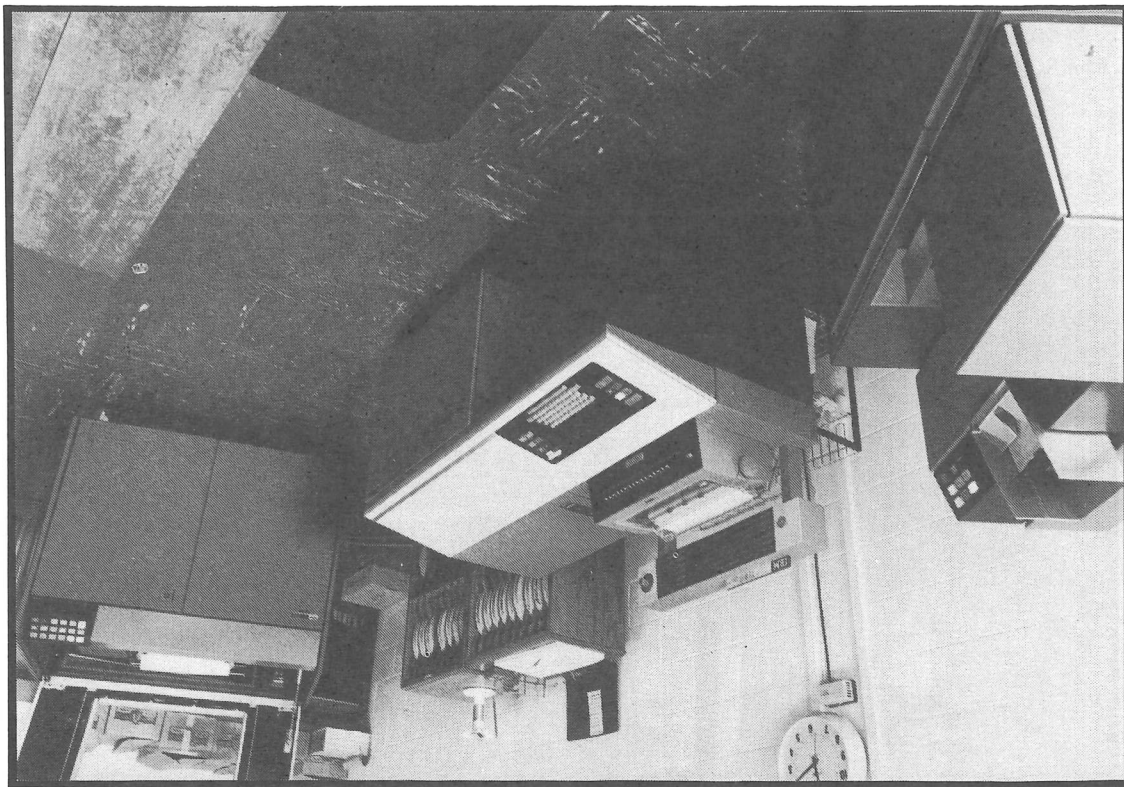
Up until this time there had been just one programmer working in Registry. Now a second position was created and Graham Weeds began in May 1982. He and Ian Johnson set about the conversion task, writing many Cobol programs, and the job went smoothly over the next few years. In 1984 the group took over responsibility for the academic side of Registry work and began to establish a student record system. Graham Weeds left Lincoln at the end of 1986, and was

replaced by Helen Instone who has continued to develop these systems. At the end of 1989 the staff numbered just two professional plus one and a half data-entry staff. They have developed and maintained a system of Cobol programs that deal with creditors, debtors, general ledger, payrolls and student records. There are four payrolls: salaries, wages, casual and student. (Student allowances are now paid fortnightly by the University). Also linked in to the financial system are a personnel system and an inventory and stock recording facility for the central store. Other facilities have been developed, the most recent being an expert system for the use of course advisors at student registration. This allows advisors to take quickly into account all regulation intricacies and the more obscure provisions for prerequisites that have sometimes escaped notice in the past.

The Library

There was very little computer use by libraries in the NZ universities prior to 1980. The Standing Committee on Library Resources arranged a meeting on library automation in May 1981 with representatives from all the university institutions. Anne Scott of the Library staff attended that meeting on behalf of Lincoln, and she came back with a proposal for immediate action for computer-produced catalogue cards. This required the adaptation of some Massey-written software (Massey were earlier in the field than most) and also the purchase of a printer with high quality output. At that time this meant a daisy-wheel printer. Both of these things were achieved by the end of 1981 and all new cataloguing was done this way. The Library was also in the process of changing its classification to Library of Congress and the existing stock of books was gradually recatalogued the same way. The catalogue was also archived on to magnetic tape in this process.

In 1984 the Serials database was computerised with software written by Andy Griffiths of the Computer Centre. Then in 1985 dial-up access was organized to the NZ Bibliographic Network that had been recently set up by National Library, followed when a telephone line became available by full-time access via a direct leased line. Instead of each library doing all its own cataloguing work, the NZBN system allowed

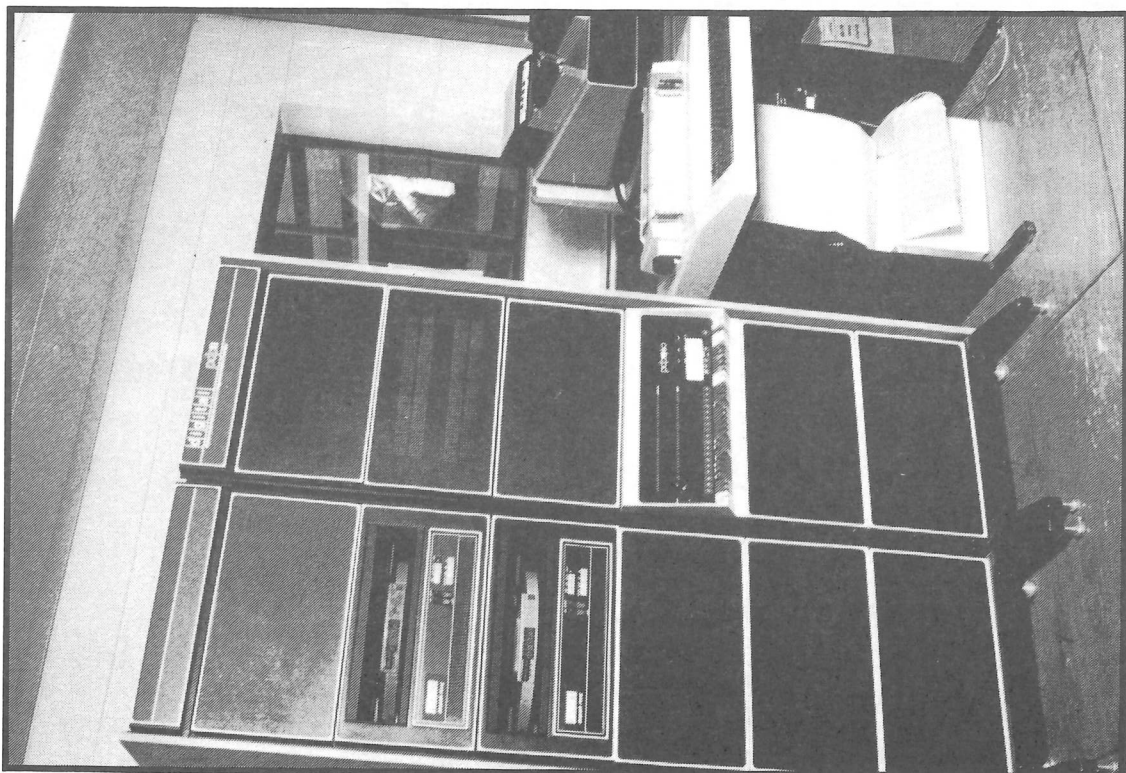


Lincoln's first computer - an IBM 1130 installed in 1976.

Delivery of the VAX 11/780 in 1981.



The PDP 11/40 installed in 1976.

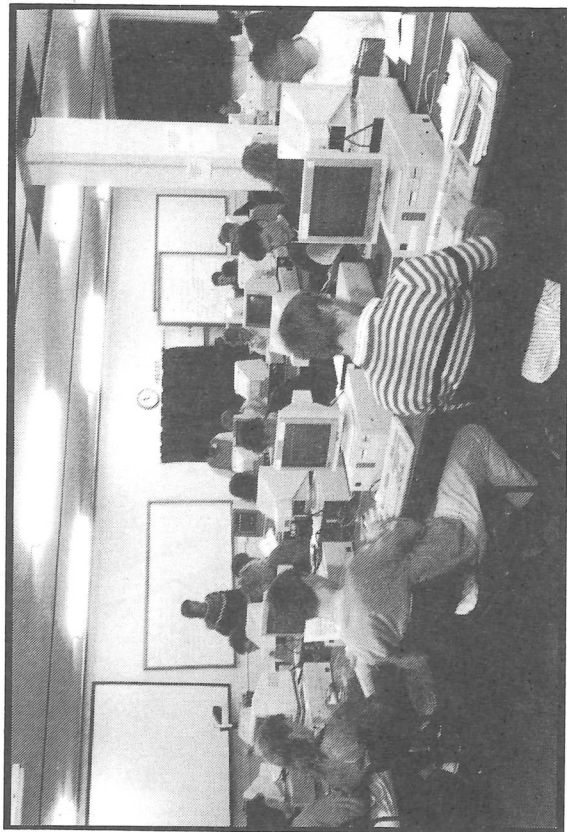


for the cataloguing of a book to be done by just one member library. Other libraries using the system could then obtain that information quickly and efficiently. Staff at the Lincoln Library could sit at a computer workstation and request the catalogue information on a new purchase. This was then sent down the line to Lincoln and catalogue cards printed directly from this. Today the information is loaded by tape onto the Library Dynix system. NZBN now provides such copy cataloguing for about 90% of the books purchased by Lincoln.

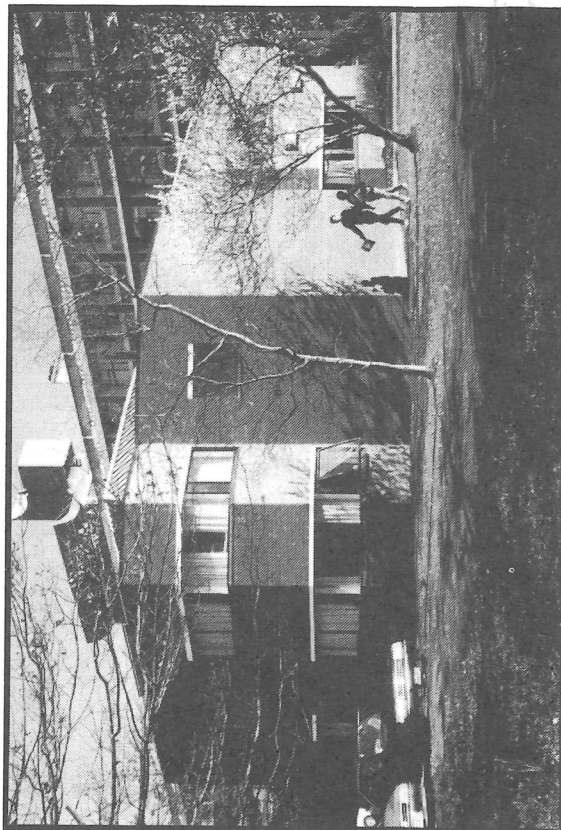
The Library shifted to a new building in October 1988 and in May 1989 installed a computerised local library system. The software was Dynix running under a Pick operating system on an Ultimate computer. Nine workstations were installed for users in the Library building and the system can be accessed from most workstations on the campus network, although the Dynix licence limits the number of simultaneous users. Another 15 workstations are used by Library staff. The system was installed with a minimum of problems. It provides online access to the whole book catalogue and students quickly learn to use it. A new development in technology is the CD-ROM for storing and accessing very large amounts of information. This is an adaptation of the audio compact disc player to provide digital information. The Library installed a CD-ROM reader in October 1989 and provide on it such things as abstracts of scientific papers. It is an area that will clearly be of increasing importance to all libraries.

The Kellogg Unit

When Peter Nuthall of the Farm Management Department was in the U.S. in the late 1970s, he saw at Purdue University a project that offered farmers a service for processing their management information. Communication between the farmer and the service was by mail, and the processing was done on a large computer. Back at Lincoln he put up a proposal for a similar service to be offered here, and the Kellogg Foundation was approached for initial funding. This was agreed to and the Kellogg Farm Management Unit was established in 1980 with Dr Nuthall as its head. This was just the time that the micro-computer was making its presence felt. There were 8-bit machines made by such



A computer teaching laboratory at Lincoln, 1990.



Lincoln Computer Centre building, with the Hilgendorf building in the background, 1990.

firms as Apple and Radio Shack being sold for what then seemed very low prices. Individuals were beginning to buy computers where previously they were bought only by organizations. Seeing the implications of this development, the unit decided to change its aim from giving a mail-order service to writing software to do the same job on the farmer's computer.

This is how its operation has continued. The first programmes were written in the BASIC language for specific machines, but soon a degree of standardization came about with the advent of the CP/M operating system which came to be used on most 8-bit machines. Later the IBM Personal Computer and its many clones provided a new standard of 16-bit machines using the Microsoft DOS operating system and programmes are provided to operate under this system too. By 1989 the Unit had about 1200 clients throughout New Zealand and Australia using packages that include a Financial Recording system, Investment Analysis, a Livestock Recording system and one on Financial Analysis. The initial period of Kellogg funding was for five years which was subsequently extended by two years. The Unit is now self-supporting with a staff of 10, some of whom are part-time.

Teaching

At the end of the 60s there was just one computing course offered at Lincoln; at the end of the 70s there were two; by 1989 nine different computing courses were available. In addition many other courses now carry some computer content. From the outset, the teaching has been aimed at educating computer users rather than pure computer scientists, and that policy continues. Lincoln students cover a wide range of topics from economics to engineering, from biochemistry to landscape architecture, from management to meteorology. All these areas have need for people of particular expertise in computing, and it is such people that Lincoln is attempting to cater for.

Since 1986 it has been possible to major in computing in the B.Com. degrees, and in 1989 118 students were taking advanced papers in computing in this degree. All commerce students do a compulsory

paper on applied computing in their first year, with an opportunity to take further papers later. Many other courses in the degree expect students to be proficient in computer use. It is a basic tool in many areas of management, finance and economics.

Agricultural science students meet computing first in the Biometrics course in their second year, and then have the option of the Experimentation course in their third year, where the use of a statistical package is central to the analysis of the field experiments that they carry out. They can take this further in their fourth year if they wish with the subject Biometrics and Computing. Again, other subjects will build on the computer training to advance in their own disciplines.

The degree in Natural Resources Engineering (formerly Agricultural Engineering) is run in conjunction with the University of Canterbury. The first two years and half the third year are spent at Canterbury, and the other half of third and all the fourth year are at Lincoln. Students do a course in Engineering Computation in their second year at Canterbury, and by the time they get to Lincoln are comfortable with the computer as a tool to work with. They are introduced to a Computer Aided Design (CAD) package at Lincoln in the third year of their course and subsequently use the computer in this and other ways for about half their assignments. In particular, they use a package for irrigation design called Irricad, which was written by staff at the Agricultural Engineering Institute on the campus. As CAD needs some specialised facilities, a laboratory of five workstations with high-resolution screens and appropriate software (Versacad) was installed in 1987 in the engineering building. It is linked to the campus network.

Lincoln also teaches degrees in Parks and Recreation Management, and in Landscape Architecture. These too have some computer content, although not as separate papers. A small CAD facility now available in the Landscape department is used extensively in their studio work. There are also some undergraduate diplomas offered at Lincoln in areas such as Farm Management and Horticultural Management. A paper in Computing has been offered in these courses since 1988.

Other Organizations

Lincoln had at this time three associated research organizations on campus: the Agricultural Economics Research Unit (AERU), the Agricultural Engineering Institute (AEI), and the Tussock Grassland and Mountain Lands Institute (TGMLI). Each of these had computing needs on a growing scale. Some of the people in the AERU had been among the first to use the Iliac computer in the 1960s and were enthusiasts from the beginnings of the Lincoln computer.

Arrangements were made for each of these organizations to contribute financially to the running of the computer. The detail of the arrangements varied according to their different needs and their different financial capabilities. It was of mutual benefit, as computing was essential for the research that was being carried on by the institutions, and the extra source of finance gave the Centre help in small hardware purchases to enhance the facilities for everyone.

There was also cooperation with some outside organizations situated in the Lincoln district. The first of these was the Wool Research Organization, which used the facilities of the Computer Centre for many years (again with a financial contribution) until they were in a position to set up their own facility. They installed a VAX of their own in 1984 and continued a network connection to the Computer Centre's VAXs. The divisions of the Department of Scientific and Industrial Research stationed in the Lincoln area always had access to computing facilities in Wellington, but also found it convenient at times to use the ones near at hand. The advent and improvement of computer network facilities reduced and then eliminated that requirement in the 1980s. In a somewhat similar way the groups of Ministry of Agriculture and Fisheries staff in the area made a good deal of use of the Computer Centre for a period. They had two terminals connected to Centre VAXs for a number of years, the first installed in 1981 and the second in 1985. In those years the Ministry had temporary accommodation on campus, but when they moved into their new buildings adjacent to DSIR they retained the link and still had one terminal connection at the end of the decade, although use had been declining for some time.

All this cooperation had spinoff advantages in the sharing of ideas and expertise. It also had an advantage for the Computer Centre when making budgetary requests. It was helpful to be able to point to a source of income outside the ordinary funding when trying to convince the Principal and Registrar of the worthiness of an item of expenditure.

The Computing Experience in the Late 1980s

Computer users in 1989 had a vastly different experience and produced very different output from their counterparts in the 1960s. Where the 1960s users mostly wrote their own programs, there was a huge array of software available to the users of 1989. Most of that software was well-written and some of it was well-documented. Many programmer-years of work went into the production of each of these software products. There were packages to do all kinds of things from word-processing to graph-making to complex computations and with a very few exceptions users found software available to do all they wanted. The 1989 user could be a landscape artist or a librarian, a secretary or an administrator, a lecturer or a student in any course. Indeed, from 1990 onwards, every student on enrolment is given a usercode on the VAX systems and PCs which comprise the campus network.

The 1989 user could sit down at a workstation connected to the campus network from which access was available to a number of computers and other facilities such as printers. The Library catalogue could be consulted, messages could be sent and received. An engineer could design by computer, a manager could plan, a research worker carry out calculations. Many of them also wrote their scientific papers at the workstation. When a researcher needed to search the journals of a discipline, this could be done too. The search need not be confined to Lincoln or even to New Zealand, because the campus network was linked to the telephone system and hence to the world. (For administrative reasons, the latter could be done by Library staff only). Staff members could collaborate with overseas colleagues on research projects via their computer workstation.

There were still occasional hassles and not everything went smoothly. Many software manuals were still hard to follow, and the degree of user-friendliness left a lot to be desired. But the scope available in 1989 was so different from that of the 1960s that computing seemed to be a totally different activity from the process of punching cards and then waiting for days for them to be processed.

Chapter 4

The Whole Period - Exponential Growth

Computing at Lincoln involves much more than the Computer Centre, as will be evident from the previous chapters. But the Computer Centre has assisted in the growth of computing in nearly all places on the campus, and on a few off campus too. It has had three names over this period - it was set up originally as the Computer Laboratory; this name was changed in 1976 to Computer Centre without any change in its structure; then in 1983 the Centre for Computing and Biometrics was established which formally brought together the service and teaching functions of computing with those of Biometrics. The Director of the Centre has always taken a full teaching workload as well as administering the service functions and the expansion of the Centre. A list of the staff who have served is printed as Appendix 1. It may be observed that a number of staff have stayed for quite long periods in an industry that is noted for the mobility of its workers. The Centre has had just two Directors in the whole period - Neil Mountier from 1967 to 1986 and Alan McKinnon from 1986 till the present. There has also been some continuity in the Chairmanship of the Computer Committee. Jim Stewart had this office from 1967 till 1974, followed by Barry Dent from 1974 till 1986, David Penman for 1986 and Tim Davies from 1987 till the present. The benefits of this continuity have outweighed any disadvantages and it has been a real factor in the success of the whole venture.

In the thirty years from the beginning of the 1960s to the end of the 1980s computing in New Zealand has come from nothing to an essential part of our lives. We depend on computers for our banking, our bills, our travel reservations, our tax demands, for the design of our buildings, for the writing of our correspondence and for hundreds of other services. The use of computers in scientific research has continued to grow. Lincoln has shared in this rapid change, and will continue to share in it, as the rate of change shows no sign of slowing. At the base of this change has been the extraordinary progress in the technologies associated with computing, so that it has become so much cheaper to use.

Charging

Growth in computing demand has often outrun the facilities available. As the cost of satisfying the demand was considerable, ways of constraining growth and rationing existing facilities were tried. University computer centres are unlike commercial centres in a number of ways, notably in having a very large number of users engaged in different tasks. Also, the distinction between work and play is not a clear one in a teaching institution, nor should it be.

The chief method used for rationing and constraining use has been to charge for it. Charging can be done in a number of ways, and the money involved can be 'real' or 'funny' money. 'Real' money can be used for another purpose if not spent on computing, 'funny' money cannot. In a 'funny' money system, the departments of the university are allocated so many dollars of computing resource, according to their estimated need. If they appear to be using that too quickly, the Computer Centre will inform them of this. If things were simple, they would be told that if they used their annual allocation in nine months then that department would be without computing for the last three months of the year. Practically, this is not possible. A research worker nearing the end of a project cannot be told that because other people in his department have been profligate, he can do no more computing for

Student Numbers

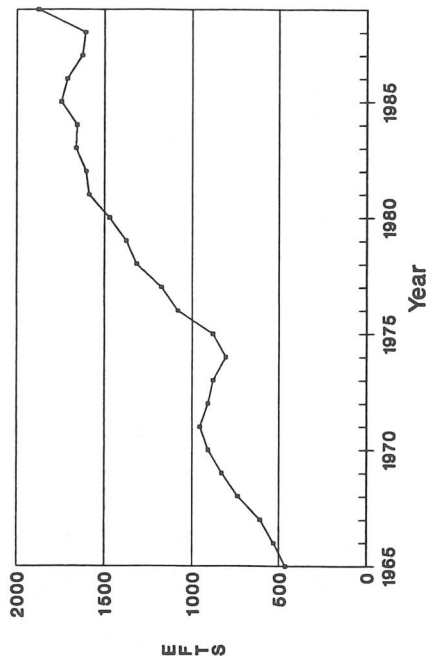


Fig 1. No. of full time students at Lincoln

Block Grant

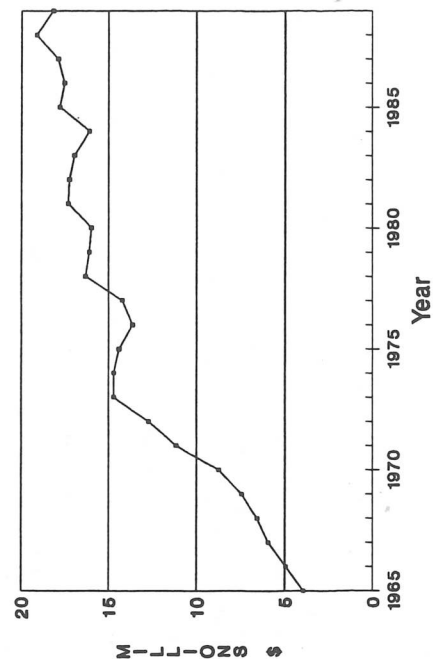


Fig 2. Lincoln Block Grant adjusted to 1988 \$

three months. Nor can a class be prevented from completing their course by such an edict. The system has to be used with compromises all the way.

This kind of charging system was used in nearly all NZ universities from the 1970s. Such a system was introduced at Lincoln in 1974, and functioned without too much strife for a number of years. In 1984 it was modified to use 'real' money for printing. This real money was credited to the Computer Centre, and its annual grant reduced in anticipation of this income. In 1988 the 'funny' money system was abandoned, and since then the only charging of internal users has been of real money for printing. Undergraduate classes are no longer included in the charging scheme.

It should be noted that all the above applies to the university's own staff and students. External users are all charged real money.

Growth of Lincoln

It would seem useful to make some specific statements about the growth and the cost of computing at Lincoln. To put this in proper context, it will be useful to look first at the growth of the institution itself. The number of students at Lincoln from 1965 to 1989 is shown in Figure 1. Equivalent Full-Time students (EFTS) is the measure used, and it can be seen that this has gone up by a factor of four in the period. The main funding of New Zealand universities comes from the Government by means of the Block Grant. Since inflation makes nonsense of raw monetary figures, these have been adjusted by the Consumer Price Index to the December 1988 dollar. Figure 2 shows the Lincoln block grant over the period 1965-89 adjusted in this way, and it can be seen that it follows a similar pattern to student numbers except for a period in the early 1970s. Apart from that period, the amount of block grant has varied round about \$10,000 per student.

With that established as a base, we consider computing growth within Lincoln. In the 1970s, when computer centre directors were engaged upon making cases for further funding for computers, some measures

of usage were quoted. In particular, the length of time that the central processor of a large computer was active was one such measure - CPU time it was called. It was easy to record this for a few large computers and it was done as a matter of course. As interactive computing became widespread, 'terminal connect time' was also recorded. This was the sum over all terminals on campus of the time that they were logged on to a computer. Yet another measure was of output - the number of pages printed. Each of these measures had deficiencies that were apparent at the time: CPU time placed an undue weight on the large 'number-crunching' jobs such as crystallography; terminal connect time included time when the user had been called away and had not logged off; and pages printed, while it seemed in some ways to give a good estimate of workload, emphasises the work of people who want everything out on paper, useful or not. All these measures were used, because some way of describing computing growth was needed. There was a rough agreement between them, so that it could be said with some confidence that computer usage appeared to double every two years, except when it was severely constrained by lack of facilities. This was not just a Lincoln experience, but was found fairly generally around NZ universities.

Computer Power

When we try to look at growth in usage today, we find that we can no longer use these measures at all, because so much computing is now done on personal computers. The operating systems of large computers provide for the recording of CPU time, terminal connect time and number of pages printed, but this is not so for PCs. The figures are just not there. It is very difficult to quantify 'computer usage' on a present-day campus. What we can attempt is to indicate the growth of computer power on campus, rather than computer usage. Two measures are offered here - the amount of memory in computers on campus and the number of workstations i.e. terminals and personal computers.

Memory

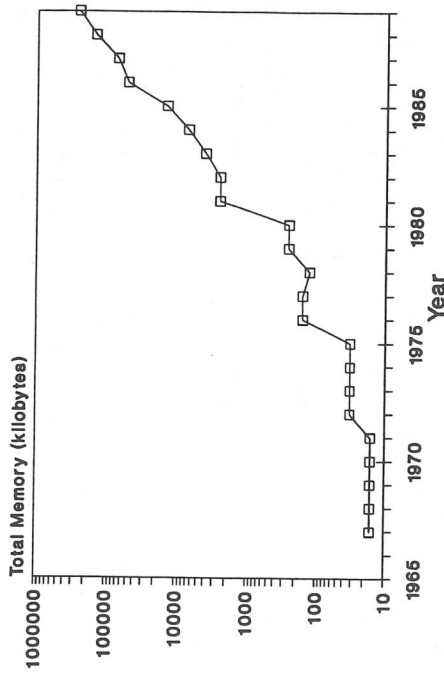


Fig 3. Total Memory of Computers installed at Lincoln

Workstations

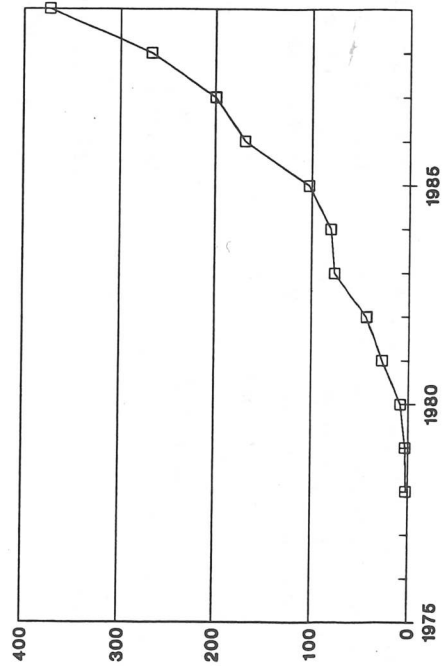


Fig 4. Number of Workstations installed at Lincoln

The amount of memory available (memory immediately available to the processor, which excludes storage on disk or tape) is shown in Figure 3. Note that the graph is on a logarithmic scale to accommodate an increase over the period by a factor of more than 10,000. (Remember that the growth in student numbers over the same period was by a factor of four). The IBM 1130 as originally installed contained memory for a little more than 16,000 characters. In the common notation of computing this is referred to as 16 kilobytes or 16KB. This was added to in the early 1970s to make 32KB. The PDP11/40 that was installed in 1976 had 128KB of main memory. When the VAX11/780 was installed in 1981, its memory was 2,500KB, more commonly known as 2.5MB (megabytes). The amount of such memory on campus at the end of 1989 totals 250MB of which 96MB is on computers in the main computer room, and the rest in personal computers of varying power around the campus. The growth in computer memory on campus over 22 years is in line with the estimates of usage growth referred to above. Doubling every two years would result in growth by a factor of 8192 in 22 years; the growth in memory is a little greater than this.

The rate of increase of workstations is less spectacular and is shown on a natural scale in Figure 4. It will be noted that there were no operative workstations until 1978, and initially growth was quite slow. Nevertheless, the graph is consistent with an average doubling every two years over the period 1979 to 1989. It is characteristic of exponential growth that it appears to start slowly and accelerate.

The Cost of Computing

Now let us look at some of the costs involved in providing this computing power. As before inflation effects are removed by quoting all money in 1988 dollars. First the actual costs of the earlier major equipment purchases are of interest. In terms of 1988 dollars the IBM 1130 installed in 1967 cost \$470,000, the PDP11/40 installed in 1976 \$220,000 and the VAX11/780 installed in 1981 \$490,000. These figures help to emphasise the boldness of the Lincoln Council in their decision to purchase the first computer in the 1960s.

Capital Expenditure

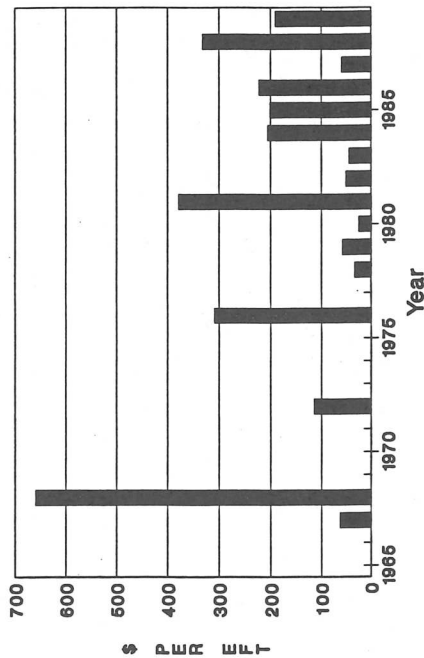


Fig 5. Computer Centre Capital Costs per student, 1988\$

Computer Centre Staff

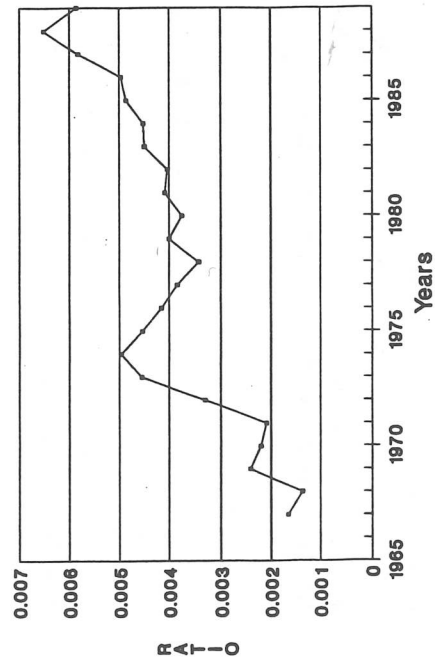


Fig 6. Ratio of Computer Centre Staff to students (EFTS)

Computer Centre Expenditure

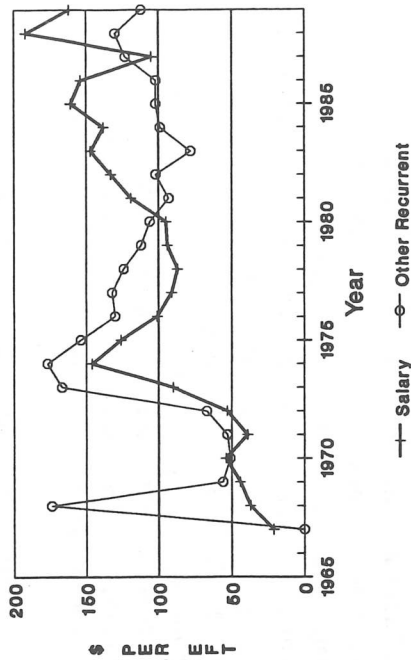


Fig 7. Running costs in 1988 \$ per student

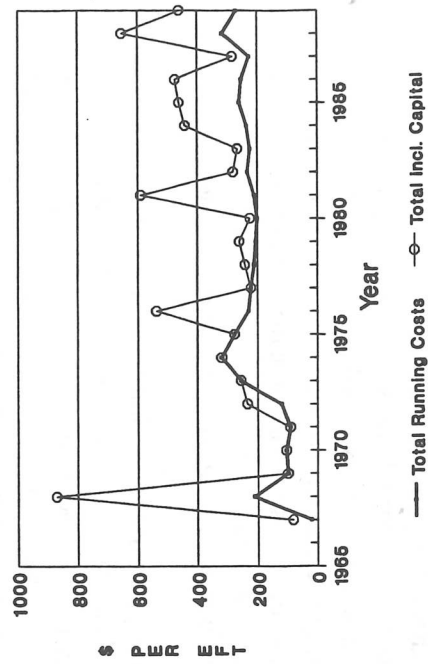


Fig 8. Total Costs in 1988 \$ per student

Perhaps the most satisfactory way of keeping a picture of expenditure in perspective in the growing institution is to relate it to the number of students. Accordingly, Figure 5 shows the expenditure on capital equipment in 1988 \$ per equivalent full-time student. It can be seen that capital expenditure is much more evenly distributed in the 1980s than it was in earlier years. It is also appreciably larger.

Running expenses fall into two categories. One of these is the expenses for which the director prepares an annual budget and covers equipment maintenance, software rental and maintenance and consumables such as paper. The other is salaries. The number of Computer Centre staff per student has risen slowly, as can be seen in Figure 6. (This excludes lecturing staff, and also excludes the Director who has always been a full-time academic as well as running the Centre). The actual expenditure on salaries is shown in Figure 7, along with the expenditure on other running expenses. It will be noted that the latter have remained surprisingly constant over a long period. Figure 8 gives the total of the two types of running expense and also the total overall with capital expenditure included.

It should be noted that all these graphs apply to the Computer Centre only, and do not include registry, library or research institutes costs. Also, the graph on memory does not make any allowance for the use made of the Burroughs B6700 at Ilam during part of the 1970s. However, the Computer Centre has been throughout the period very much the dominant feature of the computing scene at Lincoln.

It is clear that computing technology is continuing to progress at a startling rate. A university needs to make use of the technology for its own efficient operation and also to teach the use of it to its students. Lincoln made a bold investment in computing in 1967. After a difficult period in the 1970s progress continued well in the 1980s and at the beginning of the 90s the university is well placed to continue strongly in the whole computing area. New technology has been implemented quickly and the teaching of computing has both broadened and deepened in recent years. All students are expected to become

computer users and the opportunity for specialisation in applied computing is available in many courses. The service areas of registry and library are progressing well with their computer systems. The next decade will bring further technological advances, and Lincoln University is in a good position to take advantage of them.

APPENDIX 1

Staff employed in the Computer Centre and the CCB

(includes teaching staff in the CCB)

Walt Abell	Jun 89-
Robin Abrahams	Jul 80-Jan 82
David Anderson	Dec 85-Jul 89
Bruce Annabel	Nov 71-Dec 73
Ruth Arnst (earlier Taylor)	May 69-July 72 & Jan 75-Jan 78
John Baird	Jan 87-
Ann Baker	Feb 85-Dec 85
Craig Bennett	Apr 88-Jul 88
Martyn Brown	Nov 72-
Chris Calvert	Jan 87-Mar 89
Susan Clemes	Apr 78-
Cliff Collister	Feb 79-Nov 87 & Apr 89-Aug 90
Mary Dawson (earlier Thompson)	Nov 73-June 77
Karen Duffell	Apr 88-
Esmée Emerson	July 67-Aug 71
Grant English	Nov 84-Sep 87
Alan Ferguson	Jul 89-
Ruth Frampton	Feb 86-Jun 88
Hugh Frazer	Jul 88-
Sue Goodman	Feb 84-Jun 86

George Gregg	Feb 75-Jan 81
Andy Griffiths	Jul 81-Jan 84 & Feb 85-Oct 87
David Harper	Jan 74-Jan 75
Paul Helleur	Aug 86-
Julie Jenkinson (earlier Scammell)	Feb 78-Jun 80
Michael Kerrisk	Feb 83-Dec 84
Alison Lister	May 82-Jan 85
Brenda Lord (earlier Gissing)	Jan 82-
George Love	May 70-
Cathy Macken	Aug 81-Feb 86
Alan McKinnon	Apr 81-
Sheila McLaren	Nov 84-
Theresa McLennan	Feb 87-
Chris McLeod	Mar 84-Jun 85
Peter McNaughton	Jun 88-
Philippa Masters (earlier Hamel)	Dec 87-
Jenny Morris	Feb 82-Nov 84
Neil Mountier	Jun 65-Feb 86
Margaret Quince	Mar 84-
Bruce Robson	May 82-
Bill Rosenberg	Feb 88-
Richard Sedcole	Sep 86-
Valerie Sharman	Aug 72-Dec 74
Megan Sisson (earlier Smith)	Jan 82-Nov 85
Coralie Stopford (earlier Grenier)	Dec 89-
Andrew Thompson	Jul 85-Jan 87
David Vercoe	Jun 89-
Rose Voss	Jun 77-Dec 81
Richard Wood	Feb 86-Jun 86

APPENDIX 2

A description of some earlier equipment

THE IBM 1130

Installed August 1967, removed August 1977.

1131 CPU with 16KB core memory (later increased to 32 MB), and
exchangeable disk cartridge
1132 Line printer 80 lines per minute
1442 Card reader/punch

together with Card preparation equipment:

029 Card Punch
059 Verifier
082 Card sorter

THE PDP 11/40

Installed June 1976, removed September 1981.

11T40 with 128KB core memory & memory management, 2 RK05
disk drives and LA36 Decwriter console
CD11-B Card reader (1000 c.p.m.)
LP11-W Line printer, 230 l.p.m., 96 characters
LP11-V Line printer, 300 l.p.m., 64 characters

In 1978 and 1979 there were added
RK07 disk drive (28MB)
128KP MOS memory

THE VAX 11/780

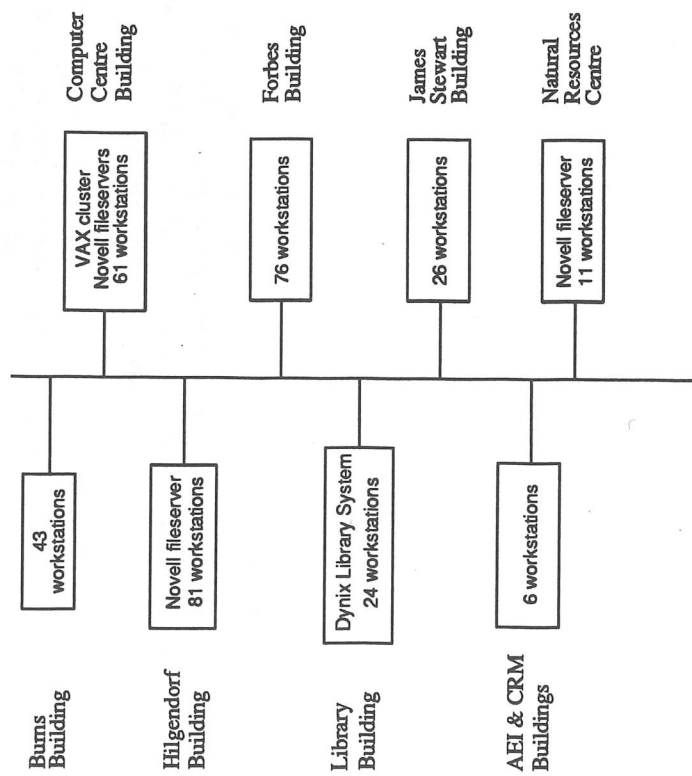
Installed April 1981, removed 1988.

11/780 with 2.5MB MOS memory
RM80 120MB disk
RM03 67MB disk
RK07 28MB disk (transferred from PDP/11)
TE16 800/1600 bpi tape drive
RX02 floppy disk drives (4)
2 Line printers (transferred from PDP/11)
32 serial lines

In 1982 an RP07 516MB disk drive was added.
Subsequently other additions and subtractions were made, with the
memory going to 4MB and finally to 8MB and the number of serial
lines being increased steadily.

APPENDIX 3

Lincoln Campus Network in December 1989



Length of backbone ethernet 1370 m

VAX cluster contains MicroVAX 3400 with 2120 Mb disk,
2 MicroVAX II's, and MicroVAX 2000